

This document describes how a customer obtains a ECHONET Subordinate CA under the SealSQ IOT ROOT GA, operated within the OISTE/WISeKey Global Trust Model (OWGTM). It has two paths: **Model A — CA operated by SealSQ (managed service)** and **Model B — CA operated by the customer**. Both are governed by the OISTE/WISeKey ECHONET CP/CPS ("CP/CPS") and by any ECHONET Requirements adopted by the PAA in CP/CPS Appendix D (the ECHONET Consortium publishes no PKI certificate policy of its own, so the CP/CPS is self-contained).

1 OVERVIEW AND ROLES

A ECHONET Subordinate CA is a Certification Authority certificate issued by the SealSQ IOT ROOT GA that authorises its holder to issue ECHONET Device Certificates for its own products. Whoever operates it, the Subordinate CA is part of the OWGTM: its policy is set by the OISTE Foundation through the Policy Approval Authority (PAA), and every Subordinate CA must be authorised by the PAA before it is issued (CP/CPS 3.2.2, 4.2.1).

Role	Responsibility in the onboarding
OISTE Foundation / PAA	Regulates the trust model and approves the CP/CPS. Two PAA representatives identify each CA Naming Application and the rightfulness of the applicant before a Subordinate CA certificate is issued from the Root (CP/CPS 4.2.1). Accepts the initial self-assessment and commissions accreditation assessments (CP/CPS 8.1, 8.4).
WISeKey / SEALSQ	Operates the SealSQ IOT ROOT GA and the INeS certificate management platform; acts as Registration Authority for the validation of the customer (CP/CPS 3.2.2, 3.2.5); performs the key ceremony and issues the Subordinate CA certificate; publishes it in the OWGTM repository, together with the CRLs of the Root CA and of each Subordinate CA (CP/CPS 2.1); in Model A hosts and operates the Subordinate CA on the customer's behalf.
Customer (Subscriber)	The legal person that will own the Subordinate CA and issue ECHONET Device Certificates for its products. Accepts the ECHONET Certificate Subscriber Agreement (v1), completes the ECHONET Subordinate CA Naming Application (v1), and in Model B additionally operates the CA under its own CPS and submits to accreditation and assessment.

2 CHOOSING THE OPERATING MODEL

The CP/CPS allows a Subordinate CA to be hosted and operated by SealSQ, or — as a technically constrained CA — by its owner (CP/CPS section 5). Other delegations or outsourcing are permitted only after a security assessment and a formal authorisation. The two models compare as follows.

Aspect	Model A — operated by SealSQ	Model B — operated by the customer
Hosting and operation	WISeKey data centres, INeS platform, SealSQ trusted roles	Customer facilities and personnel; customer CA software
CA key generation and storage	SealSQ HSM (FIPS 140-2 Level 3), witnessed ceremony (CP/CPS 6.1.1, 6.2.1)	Customer HSM, FIPS 140-2/140-3 Level 3 or equivalent, ceremony witnessed by SealSQ (CP/CPS 6.1.1, 6.2.1)
Policy documents	ECHONET Certificate Subscriber Agreement (v1); CP/CPS applies	Same, plus the customer's own CPS aligned with the CP/CPS and the consortium requirements
Technical constraint	Configured and maintained by SealSQ	Implemented by the customer, verified at accreditation and tested at every assessment (CP/CPS 7.1.5, 8.4)
Assessment	Covered by SealSQ's assessments (CP/CPS 8.4)	Initial self-assessment accepted by the PAA + OWGTM accreditation assessment before issuance; annual self-assessment and PAA-commissioned assessments thereafter (CP/CPS 8.1, 8.4)

Aspect	Model A — operated by SealSQ	Model B — operated by the customer
Deliverables from the customer	Agreement, Naming Application, commercial documents	Agreement, Naming Application, commercial documents, architecture description, CPS, self-assessment questionnaire, ceremony script and report, CSR, post-ceremony document (Sections 5 and 6)
Typical lead time	3–5 weeks from complete application to go-live	3–6 months, driven by CPS review and accreditation
Key ceremony	Conducted by SealSQ in its HSM once your inputs are complete (section 6.1)	Conducted by you in your HSM, witnessed by SealSQ by video conference; CSR signed by the Root (section 6.2)
Ongoing obligations	Section 8, commercial subscription	Section 8, plus CA operations, logging, CRL/notice services and audit access

3 PREREQUISITES COMMON TO BOTH MODELS

- You are a member of the ECHONET Consortium in good standing, or otherwise entitled by the Consortium to place ECHONET devices on the market, and hold one or more Manufacturer Codes (EPC 0x8A) assigned by the Consortium (CP/CPS 3.2.2.1).
- You are a legal person whose existence, representatives and authority can be verified (CP/CPS 3.2.2, 3.2.5).
- You accept the ECHONET Certificate Subscriber Agreement and the OISTE/WISKey ECHONET CP/CPS, and acknowledge that any adopted ECHONET Requirement prevails over both (CP/CPS 2.2.1).
- You have identified an authorised representative who can bind your organisation, and the administrators and contacts to be recorded in the Registration Authority Declaration (CP/CPS 3.2.5).
- You have concluded, or will conclude before issuance, the legally binding agreement with the OISTE Foundation required for every Subordinate CA and signed the CA Naming Request (CP/CPS 3.2.2, 4.1) — both are embodied in the ECHONET Certificate Subscriber Agreement (v1) and the ECHONET Subordinate CA Naming Application (v1).

4 MODEL A — CA OPERATED BY WISEKEY

In Model A the requisites are to accept the ECHONET Certificate Subscriber Agreement (v1) and to complete the ECHONET Subordinate CA Naming Application (v1). WISEKey does the rest.

#	Step	What happens	Owner	Output / reference
1	Initial contact and eligibility	You contact your SealSQ IoT Business Unit representative. SealSQ confirms eligibility (Section 3) and explains the service options.	Customer, SealSQ	<i>Eligibility confirmed</i>
2	Accept the agreement	You accept the ECHONET Certificate Subscriber Agreement (v1), electronically or by signing the acceptance block. This is the Subscriber Agreement of the CP/CPS.	Customer	<i>Signed agreement (CP/CPS 4.1, 4.4.1)</i>
3	Complete the Naming Application	You complete the ECHONET Subordinate CA Naming Application (v1): request details, Subordinate CA parameters and Subject DN, and sign the CA-owner approval block. Select "operated by SealSQ".	Customer	<i>Naming Application (CP/CPS 4.1 "CA Naming Request")</i>
4	Commercial set-up	Service agreement, account creation on INeS and purchase order; invoice and payment (CP/CPS 9.1).	Customer, SealSQ	<i>Service agreement, PO</i>
5	Validation	WISEKey, as Registration Authority, verifies your legal existence, ECHONET standing and identifiers, and the authority of your representatives (CP/CPS 3.2.2, 3.2.5).	WISEKey RA	<i>Validation record</i>
6	Business Unit and Trust Services approval	The IoT Business Unit Manager and the Trust Services Manager sign the Naming Application.	SealSQ	<i>Approved Naming Application</i>
7	PAA review	Two PAA representatives identify the CA Naming Application and the rightfulness to operate a Subordinate CA under the Root (CP/CPS 4.2.1).	PAA	<i>PAA approval memo</i>
8	Key ceremony and issuance	SealSQ generates the Subordinate CA key pair in its HSM and the Root signs the certificate, as described in section 6.1.	SealSQ	<i>Ceremony report, certificate (CP/CPS 6.1.1)</i>
9	Acceptance	You verify that the certificate matches your request (Subject, key fingerprint) and acknowledge receipt (CP/CPS 4.4.1).	Customer	<i>Acceptance record</i>
10	Publication	SealSQ publishes the certificate in the OWGTM repository, together with the CRLs of the Root CA and of each Subordinate CA (CP/CPS 2.1) and updates Appendix B of the CP/CPS.	SealSQ	<i>Published certificate</i>
11	INeS onboarding	Accounts, roles, API credentials and enrolment integration for the issuance of ECHONET Device Certificates; test issuance in a pre-production environment.	Customer, SealSQ	<i>Integration accepted</i>
12	Go-live	Production issuance is enabled. Ongoing obligations of Section 8 apply.	SealSQ	<i>Go-live confirmation</i>

5 MODEL B — CA OPERATED BY THE CUSTOMER

In Model B you host and operate the Subordinate CA yourself, as a technically constrained CA under CP/CPS section 5. In addition to the requisites of Model A, the CP/CPS requires you to meet the accreditation criteria of section 3.2.6, to accept the audit obligations of section 8.4, and to have your initial self-assessment accepted by the PAA before the certificate is issued (CP/CPS 3.2.2, 8.1). To evidence this you must provide the following.

5.1 Technical architecture description

A description of the environment in which the CA will operate, using the template in Appendix B. At minimum it states:

- the HSM brand, model, firmware version and certification (FIPS 140-2 or 140-3 Level 3, or Common Criteria EAL4+ against a recognised protection profile — CP/CPS 6.2.1), and whether it is operated on-premises, in a certified data centre or as a cloud HSM;
- the CA software product and version, operating system, and whether the CA is online or offline;
- the key-management design: key generation inside the HSM, m-of-n activation with at least two persons (CP/CPS 6.2.2, 6.2.8), backup and recovery of the CA key in encrypted form on HSM-bound media, and the destruction procedure (CP/CPS 6.2.4, 6.2.10);
- network architecture: segmentation, firewalls, remote-access rules and the placement of the RA/enrolment interface and of the publication services (CP/CPS 6.7);
- the physical location(s), their access controls and environmental protections (CP/CPS 5.1), and the disaster-recovery site;
- logging: what is logged, where, how integrity is protected, and how long it is retained (CP/CPS 5.4, 5.5.2);
- the time source used for certificate and log timestamps;
- how the technical constraint of CP/CPS 7.1.5 is implemented and how a request outside the constraint is rejected and logged.

5.2 Your own Certification Practice Statement

A CPS for your Subordinate CA, structured after RFC 3647 and aligned with both the OISTE/WISKey ECHONET CP/CPS and any ECHONET Requirements adopted by the PAA in CP/CPS Appendix D (the ECHONET Consortium publishes no PKI certificate policy of its own, so the CP/CPS is self-contained). Your CPS may not be less stringent than the CP/CPS on any point, and must adopt the CP/CPS certificate profiles unchanged (CP/CPS 7.1). Submit with it the alignment mapping of Appendix C, which lists, section by section, where your CPS implements each requirement. SealSQ Trust Services reviews the CPS and returns comments; issuance waits until the review is closed. Your CPS must be published in a repository you control and referenced from the certificate policy qualifier where the profile allows it.

Sections that receive particular attention in the review:

- identity validation of the devices and the source of the identifiers placed in end-entity certificates (CP/CPS 3.2.3);
- issuance controls, including the technical constraint and its testing (CP/CPS 4.2.1, 7.1.5);
- status services and compromise handling, as they apply to this hierarchy: neither ECHONET Device Certificates nor Subordinate CA certificates have a scheduled expiration (notAfter 99991231235959Z), and a retired Subordinate CA is not revoked but continues to publish its CRL while device certificates beneath it remain in service (CP/CPS 6.3.2). Revocation is by CRL only — no OCSP — with a full CRL at least every 12 months and within 24 hours of a revocation; key compromise, mis-issuance and misuse are revoked within 24 hours, other reasons within 5 days (CP/CPS 4.9.5, 4.9.7).
- trusted roles, separation of duties, background checks and training (CP/CPS 5.2, 5.3);
- audit logging, archival and retention (CP/CPS 5.4, 5.5);
- business continuity — critical services restored within 72 hours (CP/CPS 5.7) — and CA key compromise procedures (CP/CPS 5.7.3);
- CA termination and key changeover (CP/CPS 5.6, 5.8).

5.3 Self-assessment questionnaire

The completed questionnaire of Appendix A, signed by a person empowered to bind your organisation (CP/CPS 8.2). Each answer must point to evidence — a document, a configuration extract, a log sample or a screenshot — that SealSQ may ask to see during the accreditation assessment. "Partial" answers must state the remediation plan and date. The same questionnaire is used for the annual self-assessment after go-live (CP/CPS 8.1).

5.4 Recommended industry practices

The following are not all mandated by the CP/CPS but are expected of a well-run subordinate CA and will shorten the accreditation. Where you already hold the evidence, attach it to the questionnaire.

- Key ceremony: a written ceremony script reviewed by SealSQ in advance; the ceremony witnessed by SealSQ by video conference (section 6.2) and, where possible, by an independent auditor; a signed ceremony report with HSM serial numbers, key identifiers and the list of participants (CP/CPS 6.1.1).
- Offline or air-gapped operation of the CA key where the issuance volume allows it; otherwise strict network isolation of the CA host and HSM.
- Separation of duties across CA administrator, CA operator, RA officer, system administrator and security officer roles, with named individuals and dual control for all CA key operations (CP/CPS 5.2).
- Background checks and signed confidentiality undertakings for trusted roles; role-specific training before access is granted (CP/CPS 5.3).
- Centralised, tamper-evident logging (SIEM) with alerting on CA key use, configuration change and failed access; log retention for at least the period of CP/CPS 5.5.2.
- Vulnerability scanning at least quarterly and an independent penetration test of the CA and RA environment at least annually, with findings remediated or risk-accepted in writing.
- A pre-production CA and test hierarchy for integration and profile testing; automated linting of every issued certificate against the CP/CPS profile before release to production.
- A tested business-continuity and disaster-recovery plan, including recovery of the CA key from backup at the secondary site (CP/CPS 5.7).
- An incident-response procedure naming who submits Certificate Problem Reports to cps@wisekey.com and who is reachable 24x7 (CP/CPS 4.9.3, 4.9.12).
- Change management: SealSQ is informed in advance of changes to the HSM, CA software, location, trusted-role holders or the technical constraint.
- Secure manufacturing: the environment in which ECHONET Device Certificates and device keys are installed is physically and logically controlled, and key material never leaves protected storage in plaintext (CP/CPS 6.2.7, 6.4.1).
- An information security management system aligned with ISO/IEC 27001 (or equivalent), and liability insurance covering CA operations (CP/CPS 9.2).

5.5 Model B process

Steps 1 to 6 of Model A apply, with the Naming Application marked "operated by the Manufacturer / customer". The following steps replace steps 7 to 12; the key ceremony itself is described in section 6.2.

#	Step	What happens	Owner	Output / reference
7	Document submission	You submit the architecture description (App. B), your CPS with the alignment mapping (App. C) and the signed self-assessment questionnaire (App. A), with supporting evidence.	Customer	Accreditation file (CP/CPS 3.2.2, 8.1)
8	Document review	SealSQ Trust Services reviews the file and returns comments; you revise until the review is closed.	SealSQ Trust Services	Review report
9	Accreditation assessment	WISeKey, or an assessor appointed by the PAA, assesses your environment against the CP/CPS — remotely or on site — including a functional test that a request outside your technical constraint is rejected (CP/CPS 8.4).	WISeKey / PAA assessor	Assessment report
10	PAA review and approval	Two PAA representatives identify the CA Naming Application; the PAA accepts the initial self-assessment and the assessment report (CP/CPS 4.2.1, 8.1).	PAA	PAA approval memo

#	Step	What happens	Owner	Output / reference
11	Customer key ceremony	You generate the CA key pair in your HSM following the agreed script, in a video conference witnessed by SealSQ, and produce the CSR — section 6.2, steps 1 to 3.	Customer, SealSQ witness	<i>CSR, ceremony report (CP/CPS 6.1.1)</i>
12	Issuance	SealSQ verifies the CSR against the approved Naming Application and the Root signs the Subordinate CA certificate with the profile of CP/CPS 7.1.2; the certificate is returned to you — section 6.2, step 4.	SealSQ	<i>Certificate</i>
13	Acceptance and installation	You verify the certificate (Subject, fingerprint, extensions), acknowledge receipt and install it in the CA system; installation constitutes acceptance (CP/CPS 4.4.1).	Customer	<i>Acceptance record</i>
14	Post-ceremony document	You submit the post-ceremony document of Appendix D: how the CA is configured, with a sample issued certificate and, where applicable, a sample CRL — section 6.2, step 6.	Customer	<i>Post-ceremony document</i>
15	Publication	SealSQ publishes the certificate in the OWGTM repository, together with the CRLs of the Root CA and of each Subordinate CA (CP/CPS 2.1) and updates Appendix B of the CP/CPS.	SealSQ	<i>Published certificate</i>
16	Production authorisation	After review of the post-ceremony document and linting of the sample ECHONET Device Certificates, SealSQ authorises production issuance. Ongoing obligations of Section 8 apply.	SealSQ	<i>Go-live confirmation</i>

6 GENERATING THE SUBORDINATE CA: KEY CEREMONY AND CERTIFICATE ISSUANCE

Whichever model applies, the Subordinate CA key pair is generated inside a Hardware Security Module meeting CP/CPS 6.2.1, in a ceremony that follows a pre-defined script, under dual control, with an audit trail sufficient to prove that it was executed as planned, and closed by a signed ceremony report (CP/CPS 6.1.1). The private key never exists outside the HSM in plaintext. What differs between the models is who runs the ceremony and where the key lives.

6.1 Model A — ceremony conducted by SealSQ

Once you have supplied the information listed below, SealSQ schedules and conducts the ceremony; you have nothing to generate or install. The Subordinate CA is created in SealSQ's HSM and operated on the INeS platform on your behalf.

Inputs SealSQ needs from you before the ceremony is scheduled:

- the ECHONET Certificate Subscriber Agreement (v1), accepted;
- the ECHONET Subordinate CA Naming Application (v1), complete and signed, with the Subject DN, the Subordinate CA parameters and the constraint data exactly as they are to appear in the certificate and in the CA configuration — these values cannot be changed after issuance;
- the service agreement and purchase order (CP/CPS 9.1);
- the evidence of eligibility of Section 3 and of the authority of your representative (CP/CPS 3.2.2, 3.2.5);
- the name and e-mail address of the person who will verify and accept the certificate on your behalf, and whether you wish to witness the ceremony by video conference.

What SealSQ does:

- validates your application as Registration Authority and obtains the approvals of the IoT Business Unit, Trust Services and the two PAA representatives (CP/CPS 3.2.2, 4.2.1);
- prepares the ceremony script: HSM identification, participants and roles, key generation parameters (an EC P-256 or P-384 key pair), certificate profile of CP/CPS 7.1.2, verification steps;
- conducts the ceremony under dual control and under the supervision of SealSQ internal audit: the key pair is generated inside the HSM, the CSR is built from it, and the SealSQ IOT ROOT GA — an offline system — signs the Subordinate CA certificate in a Root signing session (CP/CPS 6.1.1, 6.7);
- configures the Subordinate CA on INeS with the technical constraint of CP/CPS 7.1.5 and the end-entity certificate template, and performs a test issuance;
- sends you the certificate, the chain bundle and the ceremony report, and asks you to verify the Subject, the SHA-256 fingerprint and the extensions against your application and to confirm acceptance (CP/CPS 4.4.1);
- publishes the certificate in the OWGTM repository, together with the CRLs of the Root CA and of each Subordinate CA (CP/CPS 2.1) and updates Appendix B of the CP/CPS.

Root signing sessions are scheduled events; SealSQ will give you the ceremony date once the inputs above are complete and approved.

6.2 Model B — ceremony conducted by the customer, witnessed by SealSQ

You generate the Subordinate CA key pair in your own HSM, under SealSQ's observation, and SealSQ signs the certificate from the CSR you produce. The ceremony can take place only after the accreditation file of Section 5 has been accepted and the PAA has approved the CA Naming Application (Model B steps 7 to 10).

#	Step	Description
1	Pre-ceremony preparation	You send SealSQ the ceremony script at least ten business days before the proposed date. The script identifies the HSM (brand, model, serial number, firmware — consistent with Appendix B), the participants and their trusted roles, the m-of-n activation scheme, the key generation parameters (an EC P-256 or P-384 key pair), the CSR contents, and the verification and record-keeping steps. SealSQ reviews it, agrees the date, and names its witness. You set up a video conference with a camera on the HSM and on the operator console, and a screen share of the console output.

#	Step	Description
2	Key ceremony (video-witnessed)	In the video conference, the participants identify themselves and the HSM (serial number read out and shown), confirm the HSM is in its expected state, activate it under m-of-n control, and generate the key pair inside the HSM. The private key is created as non-exportable in plaintext; where a backup is taken it is made to HSM-bound encrypted media under dual control (CP/CPS 6.2.4, 6.2.6). The public key's SHA-256 hash (the future subjectKeyIdentifier) is read out and recorded by both parties. Every action is recorded in the ceremony log with time and actor.
3	Certificate Signing Request	Still during the ceremony, you generate a PKCS#10 CSR signed with the new private key, containing the Subject DN approved in the Naming Application (Common Name, Organization, Country) and the public key. The CSR's SHA-256 fingerprint is read out on the call and noted by the witness. You transmit the CSR to SealSQ through the INeS portal or by e-mail signed by your authorised representative; SealSQ compares the fingerprint with the value recorded during the ceremony.
4	Issuance by SealSQ and return of the certificate	SealSQ verifies the CSR signature and public key, checks the requested Subject against the approved Naming Application and confirms the PAA approval memo is on file. In a Root signing session the SealSQ IOT ROOT GA signs the Subordinate CA certificate with the profile of CP/CPS 7.1.2 — extensions, policy OID, constraints and validity are set by SealSQ from the approved application, not taken from the CSR. SealSQ returns the signed certificate together with the Root certificate (the chain bundle) to your authorised representative and publishes the SHA-256 fingerprint for confirmation.
5	Installation and acceptance	You verify that the certificate's public key and subjectKeyIdentifier match the key generated at the ceremony, that the Subject and extensions are as approved, and install the certificate in the CA system; installation constitutes acceptance (CP/CPS 4.4.1). You configure the end-entity certificate template, the technical constraint of CP/CPS 7.1.5 and the status service applicable to this hierarchy, and issue a certificate to a test device.
6	Post-ceremony document	Within ten business days of issuance you submit the post-ceremony document (Appendix D), signed by your authorised representative. It records how the CA is configured — in particular CRL publication frequency and URL (full CRL at least every 12 months and within 24 hours of a revocation), the CDP/AIA values, the name constraints and the Manufacturer Code configuration — and attaches the signed ceremony report, the Subordinate CA certificate, a sample certificate issued to a test device and, where applicable, a sample CRL. SealSQ lints the samples against the CP/CPS profile and reviews the configuration; production issuance is authorised once the review is closed (Model B step 16).

Any deviation from the agreed script during the ceremony is recorded in the ceremony report with the witness's observation. A ceremony that cannot be completed as scripted is stopped and rescheduled; a key pair generated outside the witnessed session is not certified.

7 DELIVERABLES CHECKLIST

Deliverable	Model A	Model B
ECHONET Certificate Subscriber Agreement (v1), accepted	Required	Required
ECHONET Subordinate CA Naming Application (v1), completed and signed	Required	Required
Service agreement and purchase order	Required	Required
Evidence of eligibility (Section 3) and of authority of the representative	Required	Required
Technical architecture description (Appendix B), incl. HSM brand, model and certification	—	Required
Customer CPS and alignment mapping (Appendix C)	—	Required
Self-assessment questionnaire (Appendix A), signed	—	Required

Deliverable	Model A	Model B
Key ceremony script (before) and signed ceremony report (after) — section 6.2	—	Required
CSR generated in the customer HSM during the witnessed ceremony	—	Required
Post-ceremony document (Appendix D) with sample certificate(s) and, where applicable, sample CRL	—	Required
Supporting evidence for Section 5.4 practices (penetration test, BCP test, ISO 27001, insurance)	—	Recommended
Sample end-entity certificates for profile linting	Required	Required

8 OBLIGATIONS AFTER GO-LIVE

- Submit the annual self-assessment and submit to the assessments of the PAA (CP/CPS 8.1, 8.4).
- Operate the CRL service of your Subordinate CA: full CRL at least every 12 months and within 24 hours of any revocation, nextUpdate no more than 12 months ahead, and continuing after the Subordinate CA is retired for as long as device certificates issued beneath it remain in service (CP/CPS 4.9.7, 6.3.2).
- Maintain your Consortium standing and Manufacturer Code entitlement and notify WISEKey if either ceases; identity and entitlement are re-verified every 825 days (CP/CPS 3.2.2.1).
- Submit a Certificate Problem Report or revocation request through the INeS IoT Security Portal or cps@wisekey.com within the deadlines of CP/CPS 4.9.5 (CP/CPS 4.9.3, 4.9.12).
- Notify SealSQ of any change to your legal identity, contacts, Manufacturer Codes, HSM or CA architecture before it takes effect.
- Model B only: keep the CA key in the certified HSM in which it was generated, maintain the technical constraint and the configuration recorded in the post-ceremony document, retain logs and records for the period of CP/CPS 5.5.2, and grant the access required for PAA-commissioned assessments (CP/CPS 8.4). Failure to submit a self-assessment when due, or obstruction of an assessment, is a breach that can lead to suspension of issuance rights and retirement or revocation of the Subordinate CA (CP/CPS 8.4).

9 CONTACTS

Commercial and onboarding	Your SEALSQ IoT Business Unit or Sales representative
Policy, CPS review, assessments	WISEKey Trust Services — cps@wisekey.com
Certificate Problem Reports	cps@wisekey.com (CP/CPS 4.9.3)
General support	support@wisekey.com
Repository	https://github.com/OISTE/repository · https://www.wisekey.com/repository

A APPENDIX — SELF-ASSESSMENT QUESTIONNAIRE

Answer every item for the Subordinate CA you will operate. Response values: Yes (implemented and evidenced), Partial (implemented in part — state the gap and the remediation date), No, N/A (state why). Reference the evidence you can produce. Sign the declaration at the end.

#	Requirement	Response	Evidence / comment
Governance and documentation			
Q1	A CPS for the CA exists, follows RFC 3647, is approved by your management, published, and aligned with the CP/CPS (mapping in Appendix C). <i>CP/CPS 1.5, 9.12</i>	Select...	Evidence / comment
Q2	The CPS is reviewed at least annually and whenever the CP/CPS or the consortium requirements change; a change-management procedure exists. <i>CP/CPS 1.5.4</i>	Select...	Evidence / comment
Q3	Information-security policy and risk assessment cover the CA; an ISMS (e.g. ISO/IEC 27001) is in place or planned. <i>CP/CPS 6.6.2</i>	Select...	Evidence / comment
Q4	Liability insurance covering CA operations is in force. <i>CP/CPS 9.2</i>	Select...	Evidence / comment
Identity validation and registration			
Q5	Enrolment requests originate only from production systems authenticated to the CA; the RA function is documented and staffed by trained officers. <i>CP/CPS 3.2.3, 4.2.1</i>	Select...	Evidence / comment
Q6	Identifiers placed in end-entity certificates are taken from a controlled production record; uniqueness is enforced at issuance. <i>CP/CPS 3.1.5, 3.2.3</i>	Select...	Evidence / comment
Q7	Requests are checked against rejected requests and, where applicable, revoked certificates or compromise notices before issuance. <i>CP/CPS 4.1.2, 4.2.1</i>	Select...	Evidence / comment
Certificate profiles and issuance			
Q8	Certificate profiles are implemented exactly as CP/CPS 7.1.2–7.1.4; profile changes require WiSeKey approval. <i>CP/CPS 7.1</i>	Select...	Evidence / comment
Q9	Every issued certificate is linted against the profile before release; sample certificates are available. <i>CP/CPS 7.1</i>	Select...	Evidence / comment
Q10	The technical constraint of CP/CPS 7.1.5 is enforced by configuration; a test request outside the constraint is rejected and the rejection logged. <i>CP/CPS 7.1.5, 8.4</i>	Select...	Evidence / comment
Q11	Key algorithms and hashes are restricted to those of CP/CPS 6.1.5 and 7.1.3; public-key quality checks are performed. <i>CP/CPS 6.1.5, 6.1.6</i>	Select...	Evidence / comment
ECHONET-specific requirements			
Q-E1	Your CPS commits to the OISTE/WiSeKey ECHONET CP/CPS as the self-contained policy for the hierarchy and to any ECHONET Requirement adopted in Appendix D. <i>CP/CPS 2.2.1</i>	Select...	Evidence / comment
Q-E2	The Subordinate CA certificate carries the critical nameConstraints of CP/CPS 7.1.5, and the CA software is additionally configured to issue only for your Manufacturer Codes; a request for another code is rejected and logged. <i>CP/CPS 7.1.5, 8.4</i>	Select...	Evidence / comment
Q-E3	Device certificate profile: Subject C/O/OU MFRCODE/OU PRODCODE/CN Production Number, no SAN, keyUsage digitalSignature + keyAgreement, EKU clientAuth, CDP and AIA of the Subordinate CA, notAfter 99991231235959Z (no scheduled expiration) unless a fixed validity was requested for the product line. <i>CP/CPS 6.3.2, 7.1.2, 7.1.4</i>	Select...	Evidence / comment

#	Requirement	Response	Evidence / comment
Q-E4	Product Code and Production Number are taken from the production record; Production Number uniqueness within the Product Code is enforced at issuance and a duplicate is rejected. CP/CPS 3.1.5, 3.2.3.1	Select...	Evidence / comment
Q-E5	A CRL is published at least every 12 months and within 24 h of any revocation; the CRL URL in the CDP is reachable 24x7; no OCSP is offered. CP/CPS 4.9.7, 4.9.9, 4.10	Select...	Evidence / comment
Q-E6	Revocation can be executed within 24 h for key compromise, mis-issuance or misuse, and within 5 days otherwise, after investigating a Certificate Problem Report within 24 h. CP/CPS 4.9.5	Select...	Evidence / comment
Q-E7	Device private keys are generated and held in a hardware-backed key store (secure element, enclave or TPM) or, where impossible, in an encrypted container recorded and accepted by the PAA; factory injection follows CP/CPS 6.4.1. CP/CPS 6.2.7, 6.4.1	Select...	Evidence / comment
Q-E8	Key escrow is not performed and suspension is not offered. CP/CPS 4.12, 4.9.13	Select...	Evidence / comment
Facility and physical security			
Q12	The CA and HSM are housed in a secured facility with nested perimeters, access control, video surveillance and intrusion detection; access is logged. CP/CPS 5.1.1, 5.1.2	Select...	Evidence / comment
Q13	Power, air conditioning, water and fire protections are in place; media are stored in fireproof, access-controlled containers at redundant locations. CP/CPS 5.1.3–5.1.6	Select...	Evidence / comment
Q14	Off-site backups exist and restoring them requires dual control. CP/CPS 5.1.8	Select...	Evidence / comment
Personnel and trusted roles			
Q15	Trusted roles are defined (CA administrator, CA operator, RA officer, system administrator, security officer); separation of duties is enforced for CA key operations and CA certificate issuance. CP/CPS 5.2.1, 5.2.4	Select...	Evidence / comment
Q16	Background checks (references, qualifications, identity) are performed before appointment; confidentiality undertakings are signed. CP/CPS 5.3.2, 5.3.7	Select...	Evidence / comment
Q17	Role-specific training is given before access and repeated on change; sanctions for unauthorised actions are defined. CP/CPS 5.3.3, 5.3.6	Select...	Evidence / comment
Audit logging and records			
Q18	CA key lifecycle, certificate lifecycle, security and network events of CP/CPS 5.4.1 are logged with time, actor and outcome. CP/CPS 5.4.1	Select...	Evidence / comment
Q19	Logs are protected against modification, reviewed regularly, backed up, and retained for the period of CP/CPS 5.5.2. CP/CPS 5.4.3–5.4.5, 5.5.2	Select...	Evidence / comment
Q20	Registration and issuance records linking each certificate to a device and to the request are archived and retrievable for auditors. CP/CPS 5.5.1, 5.5.7	Select...	Evidence / comment
Q21	Vulnerability assessments are performed regularly and an in-depth assessment (penetration test) at least annually. CP/CPS 5.4.8	Select...	Evidence / comment
CA key management and HSM			
Q22	The CA key is generated and used only inside an HSM certified FIPS 140-2/140-3 Level 3 or CC EAL4+; brand, model and firmware are stated in Appendix B. CP/CPS 6.2.1	Select...	Evidence / comment

#	Requirement	Response	Evidence / comment
Q23	Key generation follows a written ceremony script, is witnessed, and produces a signed ceremony report. CP/CPS 6.1.1	Select...	Evidence / comment
Q24	Activation requires at least two authorised persons (m-of-n); activation data is protected and split across tokens. CP/CPS 6.2.2, 6.2.8, 6.4.1	Select...	Evidence / comment
Q25	CA key backups exist only in encrypted form on HSM-bound media under dual control; recovery has been tested. CP/CPS 6.2.4, 6.2.6	Select...	Evidence / comment
Q26	A documented key destruction procedure under dual control exists; no key archival or escrow of subscriber keys is performed. CP/CPS 6.2.5, 6.2.10, 4.12	Select...	Evidence / comment
Network and systems security			
Q27	The CA host is isolated in a dedicated network segment (or offline); administrative access is restricted, multi-factor and logged. CP/CPS 6.7, 6.5.1	Select...	Evidence / comment
Q28	Hardening, patching, malware protection and integrity checking of critical software are in place; software is verified before deployment. CP/CPS 6.5.1, 6.6.1	Select...	Evidence / comment
Q29	A trusted time source is used for certificate and log timestamps. CP/CPS 5.5.5	Select...	Evidence / comment
Status services, incidents and continuity			
Q30	Procedures exist to receive and investigate Certificate Problem Reports within 24 h and to act within the deadlines of CP/CPS 4.9.5. CP/CPS 4.9.3, 4.9.5	Select...	Evidence / comment
Q31	A CA key compromise procedure exists and names who notifies WISEKey, the consortium and relying parties. CP/CPS 5.7.3	Select...	Evidence / comment
Q32	A business-continuity and disaster-recovery plan exists, is tested, and restores critical services within 72 h. CP/CPS 5.7, 5.7.4	Select...	Evidence / comment
Q33	A CA termination and key changeover plan exists, including a successor CA and the continued CRL publication of the retired CA while device certificates beneath it remain in service. CP/CPS 5.6, 5.8, 6.3.2	Select...	Evidence / comment
Compliance and assessment			
Q34	You commit to the annual self-assessment and to PAA-commissioned assessments, including the functional constraint test, and will grant the necessary access. CP/CPS 8.1, 8.4	Select...	Evidence / comment
Q35	You will notify SealSQ in advance of changes to the HSM, CA software, location, trusted-role holders or technical constraint. CP/CPS 1.5.4	Select...	Evidence / comment

Declaration

I declare that the answers above are accurate and complete, that the evidence referenced exists and can be produced on request, and that I am empowered to bind the organisation named below (CP/CPS 8.2).

Organisation	<i>Legal name</i>	Signature
Manufacturer Code (EPC 0x8A)	<i>Six hexadecimal digits</i>	
Name and position	<i>Authorised representative</i>	
Date	<i>Select a date</i>	

B APPENDIX — TECHNICAL ARCHITECTURE DESCRIPTION (TEMPLATE)

Complete one copy per CA. Attach a network diagram and the HSM certification certificate.

CRYPTOGRAPHIC MODULE		
HSM brand and model	<i>e.g. vendor, model, form factor</i>	<i>Mandatory</i>
Firmware / software version	<i>Version in production</i>	
Certification	<i>Select...</i>	<i>CP/CPS 6.2.1; attach certificate</i>
Certificate number / protection profile	<i>NIST CMVP number or CC certificate reference</i>	
Deployment	<i>Select...</i>	
Activation (m-of-n)	<i>e.g. 2 of 5 operator cards</i>	<i>At least two persons (CP/CPS 6.2.2)</i>
Key backup and recovery	<i>Media, encryption, storage locations, dual control, last recovery test</i>	<i>CP/CPS 6.2.4, 6.2.6</i>
CA PLATFORM		
CA software and version	<i>Product and version</i>	
Operating system	<i>OS and hardening baseline</i>	
Online / offline	<i>Select...</i>	<i>CP/CPS 6.7</i>
RA / enrolment interface	<i>Protocol, authentication of production systems, rate limits</i>	<i>CP/CPS 3.2.3, 4.2.1</i>
CRL service	<i>CRL URL, issuance schedule, hosting and availability</i>	<i>CP/CPS 4.9.7, 4.10</i>
Manufacturer Code constraint	<i>How the CA software restricts issuance to your Manufacturer Codes</i>	<i>CP/CPS 7.1.5</i>
FACILITIES AND OPERATIONS		
Primary site	<i>Address, data-centre certification, physical controls</i>	<i>CP/CPS 5.1</i>
Disaster-recovery site	<i>Address and recovery time objective</i>	<i>CP/CPS 5.7 (72 h)</i>
Network architecture	<i>Segments, firewalls, remote access; diagram attached</i>	<i>CP/CPS 6.7</i>
Logging and monitoring	<i>Log sources, SIEM, integrity protection, retention</i>	<i>CP/CPS 5.4, 5.5.2</i>
Time source	<i>NTP source and stratum</i>	
Trusted roles	<i>Number of persons per role; separation of duties</i>	<i>CP/CPS 5.2</i>
Device key provisioning	<i>Where device keys are generated, how they are injected, factory controls</i>	<i>CP/CPS 6.2.7, 6.4.1</i>

C APPENDIX — CPS ALIGNMENT MAPPING (TEMPLATE)

For each section of the OISTE/WISKey ECHONET CP/CPS, state where your CPS implements the requirement and whether it is fully aligned. Add rows for consortium requirements that the CP/CPS does not restate.

CP/CPS section	Your CPS section	Alignment	Comment
1. Introduction, participants, certificate usage, policy administration	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
2. Publication and repository responsibilities	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
3.1 Naming	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
3.2 Initial identity validation	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
3.3–3.4 Re-key and revocation request identification	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
4.1–4.4 Application, processing, issuance, acceptance	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
4.5–4.8 Key usage, renewal, re-key, modification	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
4.9 Revocation / compromise handling	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
4.10–4.12 Status services, end of subscription, key escrow	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
5.1 Physical controls	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
5.2–5.3 Procedural and personnel controls	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
5.4–5.5 Audit logging and records archival	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
5.6–5.8 Key changeover, compromise and disaster recovery, termination	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
6.1 Key pair generation and installation	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
6.2 Private key protection and cryptographic module controls	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
6.3–6.4 Key pair management and activation data	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
6.5–6.8 Computer, life-cycle, network security, time-stamping	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
7. Certificate, CRL and OCSP profiles	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
8. Compliance audit and other assessments	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
9. Other business and legal matters	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>
Consortium requirements (any ECHONET Requirements adopted by the PAA in CP/CPS Appendix D (the ECHONET Consortium publishes no PKI certificate policy of its own))	<i>Your CPS §</i>	<i>Select...</i>	<i>Comment / remediation</i>

D APPENDIX — POST-CEREMONY DOCUMENT (TEMPLATE)

Model B only. Complete within ten business days of receiving the Subordinate CA certificate and return it, signed, with the attachments listed. It records the state of the CA as configured for production and is the reference against which later assessments compare the CA.

CEREMONY		
Ceremony date and location	Date, time (UTC) and site	
Participants and roles	Names and trusted roles; SEALSQ witness	CP/CPS 5.2.1
HSM identification	Brand, model, serial number, firmware; certification	Must match Appendix B
Key identifier	SHA-256 of the public key / subjectKeyIdentifier as recorded at the ceremony	
Certificate fingerprint	SHA-256 fingerprint of the Subordinate CA certificate as installed	
Deviations from the script	None, or description and witness observation	
CA CONFIGURATION		
CA software and version	As installed in production	
Key backup	Whether a backup was made, media, custodians, storage location	CP/CPS 6.2.4
End-entity certificate template	Template name; confirmation it matches CP/CPS 7.1.2 / 7.1.4	CP/CPS 7.1
Serial number generation	Random source and entropy (octets)	CP/CPS 3.1.5
CRL configuration	CRL URL, issuance schedule, nextUpdate interval (≤ 12 months), hosting and availability	CP/CPS 4.9.7, 4.10
Name constraints and Manufacturer Codes	nameConstraints present as issued; Manufacturer Codes configured; result of the rejection test	CP/CPS 7.1.5
Device certificate template	Subject attributes, notAfter 99991231235959Z or the fixed validity requested for the product line, KU/EKU, CDP and AIA of the Subordinate CA	CP/CPS 7.1.2, 7.1.4
Sample ECHONET Device Certificate	Attached: PEM of a certificate issued to a test device	CP/CPS 7.1
Sample CRL	Attached: PEM/DER of the first CRL signed by the Subordinate CA	CP/CPS 7.2
Logging	Events logged, destination, integrity protection, retention	CP/CPS 5.4
Trusted roles in production	Named holders per role; separation of duties	CP/CPS 5.2
ATTACHMENTS		
Signed ceremony report	Select...	CP/CPS 6.1.1

Subordinate CA certificate (PEM)	Select...	
Sample end-entity certificate (PEM)	Select...	Issued to a test device
Sample CRL	Select...	
Configuration extracts / screenshots	Select...	Constraint, template, status service

I confirm that the information above reflects the configuration of the CA as put into production, that the attachments are genuine, and that any later change to this configuration will be notified to SEALSQ in advance.

Organisation	Legal name	Signature
Name and position	Authorised representative	
Date	Select a date	